

# URINARY DISORDERS

PARTS OF THE URINARY TRACT ARE SUSCEPTIBLE TO INFECTIONS, RESULTING IN CONDITIONS SUCH AS CYSTITIS. SOME CHRONIC KIDNEY DISEASES ARE ALSO CAUSED BY INFECTION. COMMON SYMPTOMS, SUCH AS INCONTINENCE, CAN BE VERY TROUBLESOME.

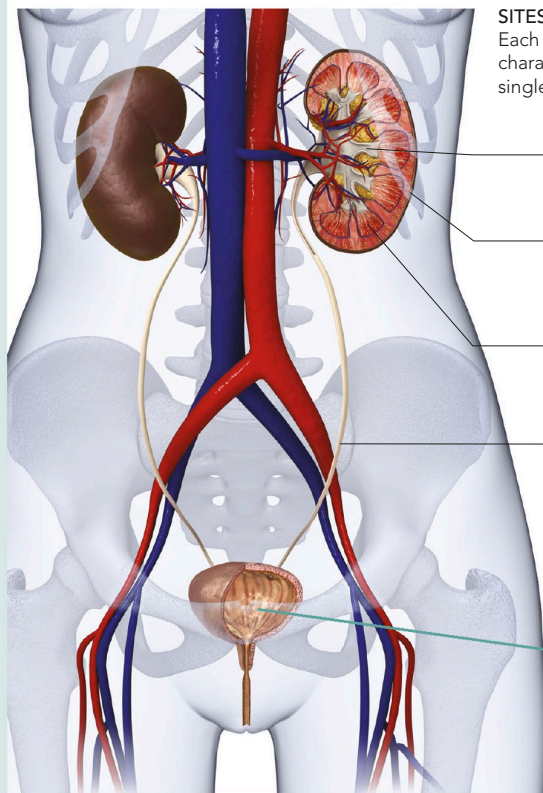
## URINARY TRACT INFECTIONS

The urine flowing through the urinary tract moves in one direction – from the kidneys through the ureters to the bladder, and then through the urethra to leave the body. During urination, the flow from the bladder is rapid and copious, but for long periods urine remains stagnant in the bladder. Infections can enter the body through the urethra and spread to the bladder, and sometimes up the ureters to the kidneys. The adult female urethra is 4cm (1½in)

long, compared to the male's at 20cm (8in). This short length and the proximity of its outlet to the anus (which means that bacteria from the anal area may enter the urethra) together account for females' greater susceptibility to urinary infection. One of the most common urinary infections is inflammation of the bladder, known as cystitis. The main symptoms of cystitis are burning pain and a frequent need to urinate, but often with little urine passed on each occasion.

### SITES OF DISORDERS

Each of the urinary organs is affected by its own characteristic diseases. However, a disorder of any single organ can affect other parts of the system.



#### Pyelonephritis

An acute infection of the urine-collecting system of the kidney

#### Diabetic nephropathy

Changes to capillaries in the kidneys, which may lead to kidney failure; caused by long-term diabetes mellitus

#### Glomerulonephritis

Inflammation of the filtering units of the kidney (glomeruli); often related to an autoimmune process

#### Reflux

Forcing of urine up the ureters by back pressure; can be caused by a blockage of the urethra

### CYSTITIS

This micrograph shows cystitis affecting a bladder lining. Bacteria (yellow rods) colonize the lining's inner surface (blue), causing inflammation.

